PADHAKU
Exploration · Class 9 Science



CHAPTER 9

Atomic Foundations of Matter

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
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🤔 Think It Over

1 **Water can be obtained from various sources. Are all these samples of water chemically identical?**

✔ ANSWER

Yes. Pure water from any source (river, borewell, ocean) is the same compound, H_2O.

It always has hydrogen and oxygen in the mass ratio **1 : 8** (Law of Constant Proportions).

Only the impurities differ; after purification all samples are identical.

2 **Oxygen is sometimes represented as O and sometimes as O_2. What is the difference between these symbols?**

✔ ANSWER

- **O** stands for one **atom** of oxygen.
- **O_2** stands for one **molecule** of oxygen, made of two oxygen atoms joined by a double bond (O=O).

Oxygen gas exists as O_2 molecules.

3 **Why does dissolved salt in water conduct electricity, but sugar does not?**

✔ ANSWER

Salt (NaCl) is an **ionic compound**. In water it gives free Na^+ and Cl^- ions, which move and carry current.

Sugar is a **covalent compound**. It dissolves as molecules and gives no ions, so it does not conduct electricity.

Activity 9.1 · Let us investigate a physical change

Activity 9.1 **Weigh a beaker with water and undissolved salt (Fig. 9.1a). Swirl till the salt dissolves and weigh again (Fig. 9.1b).**

What do you observe?

✔ ANSWER

Observation: The reading stays the same. Mass of the solution = mass of water + mass of salt.

(Tearing a paper into pieces also does not change its mass.)

Conclusion: There is no change in mass during a physical change.

Activity 9.2 · Let us investigate a chemical change

Set-up 1 · 8, 9 **Baking soda from the balloon is added to vinegar in an open conical flask. The flask and balloon are weighed again.**

What do you observe? Are the initial and the final readings same? What can be the reason for this?

✔ ANSWER

Observation: Brisk effervescence (bubbles) is seen.

The final reading is **less** than the initial reading.

Reason: Carbon dioxide gas is formed and escapes into the air from the open flask, so its mass is not weighed.

Set-up 2 · 8, 11 **The balloon with baking soda is tied to the mouth of the flask, and the baking soda is dropped into the vinegar.**

What do you observe? Are the initial and the final readings same in this case?

✔ ANSWER

Observation: Brisk effervescence; the balloon inflates as carbon dioxide collects in it.

Yes, the final reading is the same as the initial reading, because no gas escapes.

Conclusion: Total mass of reactants = total mass of products. Mass is conserved in a chemical change.

🧪 Activity 9.3 · Let us verify the law

7, 9 **Sodium sulfate solution (flask A) and barium chloride solution (flask B) are weighed together, mixed, and weighed again.**

What do you observe? Do you observe any change in the reading after mixing the solutions?

✔ ANSWER

Observation: A **white precipitate** (barium sulfate) forms.

Sodium sulfate + Barium chloride → Barium sulfate + Sodium chloride

There is **no change** in the reading after mixing.

Conclusion: Mass is conserved in a chemical reaction (Law of Conservation of Mass).

🤔 Think as a Scientist

? **Zinc + Hydrochloric acid (dilute) → Zinc chloride + Hydrogen. Design and perform an experiment to test the hypothesis that mass is conserved during this reaction.**

💡 SAMPLE ANSWER

- Pour about 10 mL of dilute hydrochloric acid into a conical flask.
- Put a few small zinc granules in a balloon and fix the balloon on the mouth of the flask with a thread (zinc should not fall in yet).
- Weigh the whole set-up and note the reading.
- Lift the balloon so the zinc falls into the acid. Bubbles of hydrogen form and the balloon swells.
- When the reaction stops, weigh the set-up again.

Expected result: Both readings are the same, so mass is conserved.

Safety: do it under teacher's guidance; hydrogen burns, so keep flames away.

⭐ Pause and Ponder · 9.1 Law of Conservation of Mass

1 **A student burns 10 g of ethanol in an open beaker. After the reaction, no residue is left in the beaker. Does this mean the Law of Conservation of Mass is violated? Explain.**

✔ ANSWER

No, the law is not violated.

Ethanol reacts with oxygen of the air and forms carbon dioxide and water vapour. Both are gases and escape from the open beaker.

Mass of ethanol + mass of oxygen used = mass of carbon dioxide + mass of water vapour formed.

If done in a closed container, no change in mass would be seen.

2 **When 20 g of hydrogen reacts completely with 160 g of oxygen, how much water is formed according to the Law of Conservation of Mass?**

✔ ANSWER

Mass of products = mass of reactants

Mass of water = 20 g + 160 g

= **180 g of water**

⭐ Pause and Ponder · 9.2 Law of Constant Proportions

3 **A compound consists of 40% sulfur and 60% oxygen by mass. In a sample of the same compound containing 20 g of sulfur, what mass of oxygen must be present?**

✔ ANSWER

Ratio sulfur : oxygen = 40 : 60 = 2 : 3

Mass of oxygen = (60 ÷ 40) × 20 g

= **30 g of oxygen**

4 **Carbon monoxide (CO) contains carbon and oxygen in the mass ratio of 3:4. How much oxygen will combine with 9 g of carbon to form carbon monoxide?**

✔ ANSWER

Carbon : oxygen = 3 : 4

Mass of oxygen = (4 ÷ 3) × 9 g

= **12 g of oxygen**

5 **The Law of Definite Proportions holds true for compounds but not for mixtures. Give reason.**

✔ ANSWER

In a compound, elements are **chemically combined** in a fixed ratio by mass, so its composition is always the same.

In a mixture, substances are only mixed, not combined. They can be present in **any ratio** (e.g., a little or a lot of sugar in water).

So the law applies only to compounds.

6 **Students X and Y prepared an oxide of copper by combining copper and oxygen in the ratios 4:1 and 8:2, respectively. Do their results justify the Law of Constant Proportions? Explain.**

✔ ANSWER

Yes.

8 : 2 = (8 ÷ 2) : (2 ÷ 2) = 4 : 1

Both students got the same ratio of copper to oxygen by mass (4 : 1). So the compound has a fixed composition, as the law says.

⭐ What if ...

? **What if atoms could combine in any ratio and not in a fixed ratio? How would this affect the substances around us?**

💡 SAMPLE ANSWER

Every sample of a substance could have a different composition. Water from two taps could be different substances.

Properties like taste, colour, boiling point and safety of medicines and food would not be fixed.

Chemical formulae could not be written, and we could not predict how much product a reaction gives.

⭐ Pause and Ponder · 9.3 Dalton's Atomic Theory

7 **Assertion (A): 2 g of hydrogen combines with 16 g of oxygen to form 18 g of water.**
Reason (R): According to Dalton's Atomic Theory, atoms combine in a simple whole number ratio by mass to form compounds.
(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

✔ ANSWER

(iii) A is true, but R is false.

A is true: 2 g + 16 g = 18 g (mass is conserved; ratio 1 : 8).

R is false: Dalton said atoms combine in a simple whole number ratio **by number of atoms** (e.g., 2 H atoms : 1 O atom in water), not by mass.

8 **Nitrogen has five valence electrons. Draw the structure of the nitrogen molecule (N_2).**

✔ ANSWER

Nitrogen (2, 5) needs 3 more electrons to complete its octet.

Each N atom shares **3 electrons** with the other N atom, so 3 pairs are shared.

Structure: **N≡N** (a triple bond). Each N also keeps one pair of its own (unshared) electrons.

Electron-dot form: :N:::N:

9 **The atomic number of fluorine is 9. Explain the formation of the fluorine molecule (F_2).**

✔ ANSWER

Fluorine: 2, 7. It has 7 valence electrons and needs 1 more.

Two fluorine atoms share **one electron each**, forming one shared pair.

Now each F has 8 electrons in its valence shell.

They are joined by a **single covalent bond**: **F—F**

10 (i) **Show the formation of carbon dioxide (CO_2).**

✔ ANSWER

C: 2, 4 (needs 4 electrons); O: 2, 6 (needs 2 electrons).

Carbon shares 2 electrons with each oxygen atom, and each oxygen shares 2 electrons with carbon.

Two double bonds are formed: **O=C=O**

Now carbon and both oxygen atoms have 8 valence electrons.

10 (ii) **Show the formation of hydrogen sulfide (H_2S).**

✔ ANSWER

H: 1 (needs 1 electron); S: 2, 8, 6 (needs 2 electrons).

Two hydrogen atoms each share one electron with one sulfur atom.

Two single bonds are formed: **H—S—H**

S gets an octet; each H gets a duplet.

10 (iii) **Show the formation of ammonia (NH_3).**

✔ ANSWER

N: 2, 5 (needs 3 electrons); H: 1 (needs 1 electron).

Three hydrogen atoms each share one electron with one nitrogen atom.

Three single bonds form: N is joined to 3 H atoms (H—N—H with the third H on N); N keeps one unshared pair.

N gets an octet; each H gets a duplet.

11 **Neon (atomic number 10) neither transfers nor shares its valence electrons. Explain.**

✔ ANSWER

Neon: 2, 8. Its valence shell already has **8 electrons** (complete octet).

It is already stable, so it does not need to lose, gain or share electrons. That is why neon exists as single atoms.

? In-text questions · 9.4.2 Bonding by electron transfer

? **Identify four elements among the first 18 elements that have less than four valence electrons and would donate them.**

✔ ANSWER

- Lithium (2, 1)
- Beryllium (2, 2)
- Sodium (2, 8, 1)
- Magnesium (2, 8, 2)

(Aluminium, 2, 8, 3, is another example.)

? **Sodium loses its one valence electron. Will it still be neutral after losing one electron? If not, what charge would it carry and why?**

✔ ANSWER

No. It becomes a positive ion, **Na^+**, with charge **+1**.

It now has 11 protons (+11) but only 10 electrons (−10), so one positive charge is left over.

⭐ What if ...

? **What if we could see atoms directly? How would it help scientists and what challenges would it cause?**

💡 SAMPLE ANSWER

Help: Scientists could watch how atoms join and break apart, check structures of molecules directly, and design new materials and medicines more easily.

Challenges: Atoms move all the time and are extremely tiny, so everything around us would look like a crowd of moving particles; ordinary objects would be hard to recognise.

⭐ Pause and Ponder · 9.4.2 Ionic bond

12 **What kind of ion will oxygen (O) form?**

✔ ANSWER

Oxygen (2, 6) gains 2 electrons to complete its octet.

It forms a negative ion (anion), the **oxide ion O^{2-}**.

13 **Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg^{2+}. However, chlorine can take only one electron to become ____. Now, ____ ion of magnesium and ____ ions of chlorine combine to give magnesium chloride.**

✔ ANSWER

Chlorine takes one electron to become **Cl^-**.

One ion of magnesium and **two** ions of chlorine combine to give magnesium chloride (**$MgCl_2$**).

14 **Show the formation of cations of potassium (K) and calcium (Ca) atoms, and the formation of their corresponding chlorides using diagrams.**

✔ ANSWER

Cations:

- K (2, 8, 8, 1) → K^+ (2, 8, 8) + e^-
- Ca (2, 8, 8, 2) → Ca^{2+} (2, 8, 8) + $2e^-$

Chloride ion: Cl (2, 8, 7) + e^- → Cl^- (2, 8, 8)

Potassium chloride: K gives 1 electron to Cl: $K^+ + Cl^-$ → **KCl**

Calcium chloride: Ca gives 1 electron each to 2 Cl atoms: $Ca^{2+} + 2Cl^-$ → **$CaCl_2$**

(Draw shells as in Fig. 9.11–9.13 of your book.)

15 **Illustrate how sodium sulfide (Na_2S) is formed.**

✔ ANSWER

Na (2, 8, 1) has 1 valence electron; S (2, 8, 6) needs 2 electrons.

Two sodium atoms each give one electron to one sulfur atom:

- $2Na$ → $2Na^+ + 2e^-$
- $S + 2e^-$ → S^{2-} (2, 8, 8)

The ions attract each other: $2Na^+ + S^{2-}$ → **Na_2S** (ionic bond).

⭐ Pause and Ponder · 9.5 Writing Chemical Formulae

16 **Name the following:**
(i) CO_2 (ii) NO_2 (iii) SF_6 (iv) PCl_3

✔ ANSWER

(i) **Carbon dioxide**

(ii) **Nitrogen dioxide**

(iii) **Sulfur hexafluoride**

(iv) **Phosphorus trichloride**

17 **Write the formula for the following:**
(i) Sodium hydrogencarbonate (ii) Sulfur dioxide (iii) Ferric chloride (iv) Cuprous oxide

✔ ANSWER

(i) **$NaHCO_3$**

(ii) **SO_2**

(iii) **$FeCl_3$** (Fe^{3+}, Cl^-)

(iv) **Cu_2O** (Cu^+, O^{2-})

18 **Write the formulae for the compounds formed from the following pairs of ions:**
(i) Fe^{3+} and OH^- (ii) K^+ and CO_3^{2-}

✔ ANSWER

(i) Charges 3 and 1 crossed → **$Fe(OH)_3$**

(ii) Charges 1 and 2 crossed → **K_2CO_3**

Activity 9.4 · Let us experiment

Table 9.2 **Test camphor, sodium chloride, copper sulfate, sugar and naphthalene for solubility in water, kerosene and petrol, and for electrical conductivity (solid and in water). Record in Table 9.2.**

✔ ANSWER

Expected observations:

Compound	Water	Kerosene	Petrol	Solid conducts?	In water conducts?
Camphor	Insoluble	Soluble	Soluble	No	No
Sodium chloride	Soluble	Insoluble	Insoluble	No	Yes
Copper sulfate	Soluble	Insoluble	Insoluble	No	Yes
Sugar	Soluble	Insoluble	Insoluble	No	No

Naphthalene	Insoluble	Soluble	Soluble	No	No

9 **Group the compounds showing similar properties listed in Table 9.2.**

✔ ANSWER

- **Group 1 (ionic):** sodium chloride, copper sulfate – soluble in water, not in kerosene/petrol; conduct electricity in water but not as solids.
- **Group 2 (covalent):** camphor, naphthalene – insoluble in water, soluble in kerosene/petrol; do not conduct.
- Sugar is covalent too: soluble in water but does not conduct.

? **Camphor and naphthalene also do not conduct electricity. Can you give a reason?**

✔ ANSWER

They are covalent compounds made of molecules, not ions. They have no charged particles that can move, so they do not conduct electricity.

? **Predict whether ionic and covalent compounds would conduct electricity in the molten state.**

✔ ANSWER

- **Ionic compounds:** Yes. On melting, ions become free to move and carry current.
- **Covalent compounds:** No. Even when melted they have no ions.

Pause and Ponder · 9.6 Properties of Ionic and Covalent Compounds

19 **What type of chemical bond is present in a solid compound that does not conduct electricity in the solid state but conducts electricity when dissolved in water?**

✔ ANSWER

Ionic bond. In the solid, ions are fixed; in water they are free to move and conduct electricity.

20 **Metal M, with two electrons in its valence shell (M shell), reacts with oxygen to form a compound that is slightly soluble in water. Predict its:**
(i) formula (ii) type of bond (iii) electrical conductivity of its aqueous solution.

✔ ANSWER

M has configuration 2, 8, 2 (magnesium). It forms M^{2+}; oxygen forms O^{2-}.

(i) Formula: **MO** (MgO)

(ii) **Ionic bond**

(iii) Its aqueous solution **conducts electricity** (weakly), because the dissolved part gives free ions.

Pause and Ponder · 9.7 Molecular Mass

21 **Find the molecular mass of nitric acid (HNO_3). Atomic mass — H = 1 u; N = 14 u; O = 16 u.**

✔ ANSWER

Molecular mass of HNO_3 = (1 u × 1) + (14 u × 1) + (16 u × 3)

= 1 u + 14 u + 48 u

= **63 u**

22 **Find the molecular mass of methane (CH_4). Atomic mass — C = 12 u; H = 1 u.**

✔ ANSWER

Molecular mass of CH_4 = (12 u × 1) + (1 u × 4)

= 12 u + 4 u

= **16 u**

⭐ Pause and Ponder · 9.8 Formula Unit Mass

23 **Find the formula unit mass of potassium chloride (KCl). Atomic mass — K = 39 u; Cl = 35.5 u.**

✔ ANSWER

Formula unit mass of KCl = (39 u × 1) + (35.5 u × 1)

= **74.5 u**

24 **Find the formula unit mass of magnesium hydroxide, $Mg(OH)_2$. Atomic mass — Mg = 24 u; O = 16 u; H = 1 u.**

✔ ANSWER

Formula unit mass of $Mg(OH)_2$ = (24 u × 1) + {(16 u × 1) + (1 u × 1)} × 2

= 24 u + 17 u × 2

= 24 u + 34 u

= **58 u**

⭐ Revise, Reflect, Refine

1 **Element A has one electron in its third shell. Element B has six electrons in its second shell.**

(i) How many electrons does A tend to give or take to become stable?

(ii) What kind of ion would it form?

(iii) How many electrons does B tend to give or take to become stable?

(iv) What kind of ion would it form?

(v) If A and B were to combine, what kind of bond would be formed?

(vi) What would be the formula for the compound thus formed?

✔ ANSWER

A = 2, 8, 1 (sodium); B = 2, 6 (oxygen).

(i) A **gives 1 electron**.

(ii) A positive ion (cation) with charge +1: $\mathbf{A^+}$

(iii) B **takes 2 electrons**.

(iv) A negative ion (anion) with charge −2: $\mathbf{B^{2-}}$

(v) **Ionic bond** (electrons are transferred from A to B).

(vi) Charges 1 and 2 crossed → $\mathbf{A_2B}$ (like Na_2O).

2 **An element X has six electrons in its outer shell and forms a diatomic molecule.**

(i) Why would that be so?

(ii) What kind of bond would it form?

(iii) Draw the structure of the molecule it would form.

(iv) Element Y has two electrons in its second shell. Draw the structure of the molecule that X would form with Y.

💡 SAMPLE ANSWER

(i) X needs 2 more electrons for an octet. Two X atoms share 2 electrons each, so both complete their octets and form X_2.

(ii) A **covalent bond** – a **double bond** (two shared pairs).

(iii) **X=X** (like O=O); each X atom also keeps two unshared pairs.

(iv) Y is 2, 2 – it has 2 valence electrons (less than four), so it gives them to X:

- $Y \rightarrow Y^{2+} + 2e^-$ and $X + 2e^- \rightarrow X^{2-}$
- $Y^{2+} + X^{2-} \rightarrow$ **YX** (e.g., BeO if X is oxygen and Y is beryllium)

3 **You want to design a new ionic compound where the total positive charge is 6+ and the total negative charge is 6−. Which combination gives the correct number of ions?**

(i) 2 Al^{3+} and 3 Cl^-

(ii) 3 Mg^{2+} and 1 PO_4^{3-}

(iii) 2 Fe^{3+} and 3 O^{2-}

(iv) 3 Ca^{2+} and 2 SO_4^{2-}

✔ ANSWER

(iii) 2 Fe^{3+} and 3 O^{2-}

2 × (+3) = +6 and 3 × (−2) = −6.

Others: (i) +6, −3; (ii) +6, −3; (iv) +6, −4 – not balanced.

4 **Choose the correct statement(s) and correct the false statement(s).**

(i) Elements are made up of molecules and compounds are made up of atoms.

(ii) The molecule of a compound is always made up of two or more atoms of the same kind.

(iii) One molecule of nitrogen gas contains three nitrogen atoms.

(iv) Water is made of two hydrogen atoms, covalently bonded with one oxygen atom.

✔ ANSWER

(i) **False.** Correct: Elements are made up of atoms of only one kind; compounds are made up of atoms of two or more different elements combined together.

(ii) **False.** Correct: The molecule of a compound is always made up of atoms of **different** kinds.

(iii) **False.** Correct: One molecule of nitrogen gas (N_2) contains **two** nitrogen atoms.

(iv) **True.**

5 **Write the chemical formulae for the following compounds.**
(i) Aluminium nitrate (ii) Calcium oxide (iii) Ferric oxide

✔ ANSWER

(i) Al^{3+}, NO_3^- → **$Al(NO_3)_3$**

(ii) Ca^{2+}, O^{2-} → **CaO**

(iii) Fe^{3+}, O^{2-} → **Fe_2O_3**

6 **Write the formulae of the compounds formed from the following pairs of ions.**
(i) Ca^{2+} and Br^- (ii) Al^{3+} and CO_3^{2-} (iii) K^+ and SO_4^{2-} (iv) NH_4^+ and Cl^-

✔ ANSWER

(i) **$CaBr_2$**

(ii) **$Al_2(CO_3)_3$**

(iii) **K_2SO_4**

(iv) **NH_4Cl**

7 **Which of the following, in Fig. 9.18, correctly represents Cl^- ion (atomic number of chlorine = 17)?**

✔ ANSWER

(ii) – it shows 2, 8, 8 electrons.

Cl has 17 electrons (2, 8, 7). Cl^- gains one electron, so it has 18 electrons: **2, 8, 8**.

(iv) shows 2, 8, 7 (neutral Cl atom); (i) and (iii) have wrong numbers.

8 **Determine the formula unit mass of the following substances.**

(i) Ammonium nitrate (NH_4NO_3) (ii) Phosphoric acid (H_3PO_4) (iii) Sodium hydrogencarbonate ($NaHCO_3$)

✔ ANSWER

Atomic masses: H = 1 u, C = 12 u, N = 14 u, O = 16 u, Na = 23 u, P = 31 u

(i) NH_4NO_3 = (14 × 2) + (1 × 4) + (16 × 3) = 28 + 4 + 48 = **80 u**

(ii) H_3PO_4 = (1 × 3) + (31 × 1) + (16 × 4) = 3 + 31 + 64 = **98 u**

(iii) $NaHCO_3$ = 23 + 1 + 12 + (16 × 3) = 23 + 1 + 12 + 48 = **84 u**

9 **Write the formulae for the compounds formed by the reaction of:**

(i) Magnesium and nitrogen (ii) Lithium and nitrogen (iii) Sodium and sulfur (iv) Aluminium and oxygen

✔ ANSWER

(i) Mg^{2+}, N^{3-} → **Mg_3N_2**

(ii) Li^{+}, N^{3-} → **Li_3N**

(iii) Na^{+}, S^{2-} → **Na_2S**

(iv) Al^{3+}, O^{2-} → **Al_2O_3**

10 **Complete Table 9.3 by writing the formulae of the compounds formed by the cations on the left (NH_4^+, Li^+, Al^{3+}, Cu^{2+}) and the anions at the top (NO_3^-, SO_4^{2-}, PO_4^{3-}). $LiNO_3$ is given as an example.**

✔ ANSWER

Cation	NO_3^-	SO_4^{2-}	PO_4^{3-}
NH_4^+	NH_4NO_3	$(NH_4)_2SO_4$	$(NH_4)_3PO_4$
Li^+	$LiNO_3$	Li_2SO_4	Li_3PO_4
Al^{3+}	$Al(NO_3)_3$	$Al_2(SO_4)_3$	$AlPO_4$
Cu^{2+}	$Cu(NO_3)_2$	$CuSO_4$	$Cu_3(PO_4)_2$

11 **5.3 g of sodium carbonate and 6.0 g of acetic acid react to produce 2.2 g of carbon dioxide, 0.9 g of water, and 8.2 g of sodium acetate. Verify whether the law of conservation of mass is valid.**

✔ ANSWER

Total mass of reactants = 5.3 g + 6.0 g = 11.3 g

Total mass of products = 2.2 g + 0.9 g + 8.2 g = 11.3 g

Mass of reactants = mass of products

So, **the law of conservation of mass is valid**.

12 **A species has 11 protons, 12 neutrons and 10 electrons.**

(i) What is its atomic number and mass number?

(ii) Is it neutral, a cation or an anion? Explain.

(iii) Write its electronic configuration.

(iv) Name the species.

✔ ANSWER

(i) Atomic number = protons = **11**; mass number = 11 + 12 = **23**

(ii) **Cation.** It has 11 positive protons but only 10 electrons, so it has a charge of +1.

(iii) **2, 8**

(iv) **Sodium ion, Na^+**

13 **Two elements A and B have configurations A: 2, 8, 5 and B: 2, 8, 7.**

(i) Which element is more reactive?

(ii) Will A and B form ionic or covalent bonds when they combine? Explain.

(iii) Predict the formula of the compound they would form.

✔ ANSWER

A is phosphorus (15), B is chlorine (17).

(i) **B** is more reactive. It needs only 1 electron to complete its octet, while A needs 3.

(ii) **Covalent bonds.** Both need electrons (A needs 3, B needs 1) and neither can easily lose 5 or 7 electrons, so they **share** electrons. A shares one electron with each of three B atoms.

(iii) **AB_3** (PCl_3)

14 **Assertion (A): Copper sulfate conducts electricity in the molten state but not in the solid state.**

Reason (R): Copper and sulfate ions are fixed in the lattice in molten state, while in solid state they can move freely.

(i) Both A and R are true, and R is the correct explanation of A.

(ii) Both A and R are true, but R is not the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

✔ ANSWER

(iii) A is true, but R is false.

R is the wrong way round: in the **solid** state ions are fixed, and in the **molten** state they move freely and conduct electricity.

15 **The species ^{27}Al, $^{80}Br^-$ and $^{201}Hg^{2+}$ have 13, 35 and 80 protons, respectively. How many electrons and neutrons do they have?**

✔ ANSWER

Neutrons = mass number − protons. Electrons: neutral atom = protons; anion adds, cation removes electrons.

Species	Electrons	Neutrons
^{27}Al	13	27 − 13 = 14
$^{80}Br^-$	35 + 1 = 36	80 − 35 = 45
$^{201}Hg^{2+}$	80 − 2 = 78	201 − 80 = 121

⭐ The Journey Beyond

1 **Design and perform an experiment to show and compare that water always contains hydrogen and oxygen in the same ratio, regardless of its source.**

💡 SAMPLE ANSWER

Collect water from 2–3 sources (tap, rain, river) and purify it (filter, boil and condense).

Under the teacher's guidance, electrolyse equal amounts of each sample with a little salt, collecting gases in inverted test tubes.

In every sample, the volume of hydrogen is about **twice** the volume of oxygen, showing water has the same composition (1 : 8 by mass) from every source.

2 **Compare atoms and ions of any three elements. Show the number of electrons before and after ion formation using bar graphs.**

💡 SAMPLE ANSWER

Element	Electrons in atom	Ion	Electrons in ion
Sodium	11	Na^+	10
Magnesium	12	Mg^{2+}	10
Chlorine	17	Cl^-	18

Draw two bars for each element (atom and ion) using these numbers.

3 **Make a card game with cations and anions.**

💡 SAMPLE ANSWER

Make cards for ions such as Na^+, K^+, Ca^{2+}, Al^{3+}, NH_4^+, Cl^-, O^{2-}, OH^-, SO_4^{2-}, CO_3^{2-}.

Players pick cards and form neutral compounds whose charges add up to zero, e.g., $Ca^{2+} + 2\ Cl^- = CaCl_2$.

Score 1 point for each correct formula and name.

⭐ The Quest Continues ...

? Are there any chemical changes that do not obey the Law of Conservation of Mass?

💡 **SAMPLE ANSWER**

In ordinary chemical reactions, mass is always conserved (any tiny change is too small to measure).

In **nuclear reactions** (which are not chemical reactions), a small amount of mass changes into a large amount of energy, so the mass of products is slightly less than that of reactants.

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**